Laws of Exponents

Algebra Worksheet · Grade 8-10

Name: ______________________ Date: ______________ Score: / 10

Learning Objectives

- Apply the seven laws of exponents to simplify algebraic expressions
- Distinguish between adding and multiplying exponents in different contexts
- Evaluate expressions involving zero, negative, and positive exponents

Simplify each expression using the laws of exponents and write your answer in simplest form.

1. Simplify using the zero exponent rule.

$(5x)^0$

Answer: ______________________

2. Simplify using the product rule of exponents.

$2x \cdot x$

Answer: ______________________

3. Simplify using the product rule of exponents.

$m^3 \cdot m$

Answer: ______________________

4. Simplify using the product rule of exponents.

$x^2 \cdot x^5$

Answer: ______________________

5. Simplify using the quotient rule of exponents.

$\frac{x^6}{x^2}$

Answer: ______________________

6. Simplify using the power of a power rule.

$(x^2)^5$

Answer: ______________________

7. Simplify using the power of a product rule.

$(5x)^2$

Answer: ______________________

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8. Rewrite using the negative exponent rule.

2^{-1}

Answer: ____________________________________

9. Evaluate using the rule for exponents of one.

7^{1}

Answer: ____________________________________

10. Simplify by combining the power of a product and power of a power rules.

$(2x^{3})^{4}$

Answer: ____________________________________

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Remind students to distinguish between the product rule (add exponents) and the power-of-a-power rule (multiply exponents).

Solutions

1. Simplify using the zero exponent rule.

$(5x)^0$

→ *Any nonzero expression raised to the zero power equals one.*
→ *Therefore the entire expression simplifies to 1.*

Answer: 1

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2. Simplify using the product rule of exponents.

$2x \cdot x$

→ *Both x terms have an exponent of one.*
→ *Add the exponents: one plus one equals two.*
→ *The coefficient 2 remains, giving 2x squared.*

Answer: $2x^2$

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3. Simplify using the product rule of exponents.

$m^3 \cdot m$

→ *The variable m has an implied exponent of one.*
→ *Add the exponents: three plus one equals four.*
→ *The simplified expression is m to the fourth power.*

Answer: m^4

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4. Simplify using the product rule of exponents.

$x^2 \cdot x^5$

→ *Since the bases are the same, add the exponents.*
→ *Two plus five equals seven.*
→ *The simplified expression is x to the seventh power.*

Answer: x^7

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5. Simplify using the quotient rule of exponents.

$\frac{x^6}{x^2}$

→ *Subtract the denominator exponent from the numerator exponent.*
→ *Six minus two equals four.*
→ *The simplified expression is x to the fourth power.*

Answer: x^4

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6. Simplify using the power of a power rule.

$(x^2)^5$

→ *When a power is raised to another power, multiply the exponents.*
→ *Two times five equals ten.*
→ *The simplified expression is x to the tenth power.*

Answer: x^{10}

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7. Simplify using the power of a product rule.

$(5x)^2$

→ *Distribute the exponent of two to both the 5 and the x.*
→ *Five squared is twenty-five and x squared remains.*
→ *The simplified expression is 25 x squared.*

Answer: $25x^2$

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8. Rewrite using the negative exponent rule.

2^{-1}

→ *A negative exponent means take the reciprocal of the base.*
→ *Two to the negative one becomes one over two.*
→ *The simplified value is one half.*

Answer: $\frac{1}{2}$

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9. Evaluate using the rule for exponents of one.

7^1

→ *Any number raised to the first power equals itself.*
→ *Therefore seven to the first power is seven.*

Answer: 7

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10. Simplify by combining the power of a product and power of a power rules.

$(2x^3)^4$

→ *Distribute the exponent four to both the 2 and the x cubed.*
→ *Two to the fourth power equals sixteen.*
→ *Multiply the exponents three and four to get x to the twelfth power.*
→ *The simplified expression is sixteen x to the twelfth.*

Answer: $16x^{12}$

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